

Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

Joseph White
Advisor: Dr. John Bowblis

Miami University

March 14, 2026

Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common:

Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common:
 - (i) Mergers and acquisitions

Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common:
 - (i) Mergers and acquisitions
 - (ii) New management

Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common:
 - (i) Mergers and acquisitions
 - (ii) New management
- This second type of change does **not** alter the set of providers in a market.

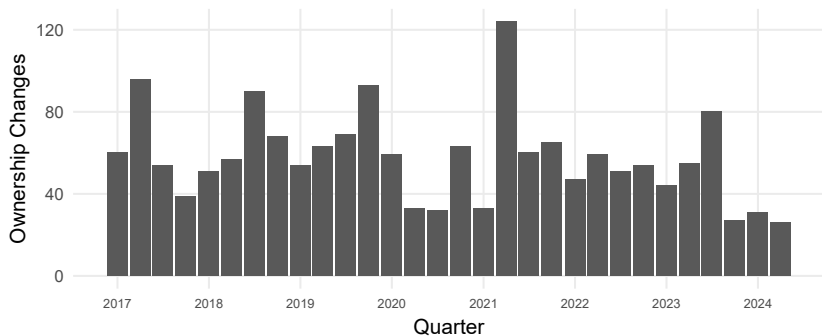
Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common:
 - (i) Mergers and acquisitions
 - (ii) New management
- This second type of change does **not** alter the set of providers in a market.
- Instead, effects operate through **internal production decisions**.

Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common:
 - (i) Mergers and acquisitions
 - (ii) New management
- This second type of change does **not** alter the set of providers in a market.
- Instead, effects operate through **internal production decisions**.
- This type of change is understudied in the literature, and we examine it in **nursing homes**.

Ownership Changes in Nursing Homes by Quarter



Why Nursing Homes?

- 15,000 U.S. nursing homes serve over 4 million people annually
- 56% of retirement age adults will use a nursing home
- Labor is the main input into care
 - ▶ More staffing improves measured quality ([Lin, 2014](#)) and decreases deficiency citations ([Harrington et al., 2000](#))
 - ▶ Major cost
- Staffing can easily change so it may adjust following an ownership change

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
 - ▶ Do responses differ across staff type?

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
 - ▶ Do responses differ across staff type?
 - ▶ Do responses differ if a nursing home is part of a chain or not?

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
 - ▶ Do responses differ across staff type?
 - ▶ Do responses differ if a nursing home is part of a chain or not?
 - ▶ Do responses differ pre vs post-pandemic?

Nursing Staff Types

- **Three main nursing roles:**

- ▶ **Registered Nurses (RNs):** planning and assessment, supervision
- ▶ **Licensed Practical Nurses (LPNs):** routine clinical tasks
- ▶ **Certified Nursing Assistants (CNAs):** direct daily care

Nursing Staff Types

- **Three main nursing roles:**
 - ▶ **Registered Nurses (RNs):** planning and assessment, supervision
 - ▶ **Licensed Practical Nurses (LPNs):** routine clinical tasks
 - ▶ **Certified Nursing Assistants (CNAs):** direct daily care
- **Staffing measure: Hours Per Resident Day (HPRD)**
 - ▶ How much total time nurses spend caring for one resident over a 24-hour period
- Ownership change can affect both **levels** and **mix** which sets up cost-quality trade-off.

Cost–Quality Trade Off

- Owners can raise quality through two channels:
 - ▶ **Quantity:** increasing total nursing hours (HPRD)
 - ▶ **Skill mix:** shifting hours toward more clinically trained staff
- **Both margins raise costs** because nursing labor is a major operating expense.
- Short-run revenues are fixed.
- **Implication:** an ownership change that shifts objectives or constraints can move a facility along this tradeoff.

Why Staffing Could Rise or Fall

- Ownership changes can shift **constraints** and **objectives** $\Rightarrow$ staffing effects are ambiguous.
- **Scenario 1: Cost pressures**
 - ▶ Debt & tighter targets $\Rightarrow$ cut labor costs
- **Scenario 2: Quality focus**
 - ▶ Invest in staffing $\Rightarrow$ increase staffing
- **Scenario 3: Reorganization**
 - ▶ Shift RN/LPN/CNA composition; total hours may be similar

Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**
 - ▶ Annual cost report information
 - ▶ Dates of ownership change
- **Nursing Home Compare (NHC) Archive**
 - ▶ Contains ownership information as well as facility characteristics
- **Payroll Based Journal (PBJ)**
 - ▶ Payroll-reported nurse staffing hours by staff type
 - ▶ Used to construct outcome variables
- **Facility-month panel, 48 contiguous states, January 2017 – June 2024**

Identifying Ownership Changes

- **Challenge:** ownership is often **layered**, so an administrative “change” may not reflect a true **transfer of economic control**.
- We identify ownership changes using a **3-step process**.

Identifying Ownership Changes

- **Challenge:** ownership is often **layered**, so an administrative “change” may not reflect a true **transfer of economic control**.
- We identify ownership changes using a **3-step process**.
- **Step 1:** HCRIS includes a variable indicating **whether and when** an ownership change occurred.
 - ▶ Useful for identifying **candidate events**, but the flag may be incomplete or capture legal restructuring rather than an economic change.
- Therefore, we have to use **NHC ownership shares** to verify and date whether control actually changes.

Ownership Level Challenge

- **NHC**

- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes

- **Step 2: Validate & date using NHC**

- ▶ Track shares
- ▶ **Majority control (50%)** transfers between $t - 1$ and t
- ▶ Do not flag name changes or family transfers

Identifying Ownership Changes

- **Step 2: Validate & date using NHC**

- ▶ Track shares
- ▶ **Majority control (50%)** transfers between $t - 1$ and t
- ▶ Do not flag name changes or family transfers

- **Example from the data:**

- ▶ *Before:*

- ★ ESTES JAMES (84%)
- ★ BURCHFIELD JOHN (10%)
- ★ LEE CLAUDE (6%)

- ▶ *After:*

- ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
- ★ JENNIFER LEE ESTES TR 031093 (50%)

- ▶ We do **not** treat this as a meaningful ownership change

Ownership Change Agreement

- **Step 3: Validate between HCRIS and NHC.**
- We keep facilities only if:
 - ▶ 0 changes in **both** NHC and HCRIS, or
 - ▶ Exactly 1 change in **each** dataset, within **6 months**.
- **Event month:** NHC date of ownership change.

Cross-tab of NHC & HCRIS

Count of changes in NHC	2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
	1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
	0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)
		0	1	2+
		Count of changes in HCRIS		

Dependent Variables: Nurse Staffing (PBJ) and Sample

- We study four staffing measures:
 - ▶ RN, LPN, CNA, and total HPRD
- Outcome is **hours per resident day (HPRD)**: staffing intensity standardized by resident census

$$\text{HPRD}_{s,i,t} = \frac{\text{TotalHours}_{s,i,t}}{\text{ResidentDays}_{i,t}}$$

Dependent Variables: Nurse Staffing (PBJ) and Sample

- We study four staffing measures:
 - ▶ RN, LPN, CNA, and total HPRD
- Outcome is **hours per resident day (HPRD)**: staffing intensity standardized by resident census

$$\text{HPRD}_{s,i,t} = \frac{\text{TotalHours}_{s,i,t}}{\text{ResidentDays}_{i,t}}$$

- **CMS implausible staffing exclusions:**
 - ▶ RN HPRD = 0 **and** LPN HPRD = 0
 - ▶ Total HPRD < 1.5 or > 12
 - ▶ CNA HPRD > 5.25
- **Additional restrictions:** beds < 15; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 treated ($\approx 20\%$)

Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions
([Goodman-Bacon, 2021](#))
- **Variation:** within-facility staffing changes relative to facilities not yet treated
- **Event study regression:**

Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions (Goodman-Bacon, 2021)
- **Variation:** within-facility staffing changes relative to facilities not yet treated
- **Event study regression:**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event_time_{it} = \tau\} \cdot Treat_i + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- ▶ Y_{it} : staffing outcome in **HPRD**
- ▶ β_{τ} : effect on Y_{it} at event time τ
- ▶ X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- ▶ α_i facility FE; λ_t month FE
- ▶ SEs clustered at the facility and month level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD regression:**

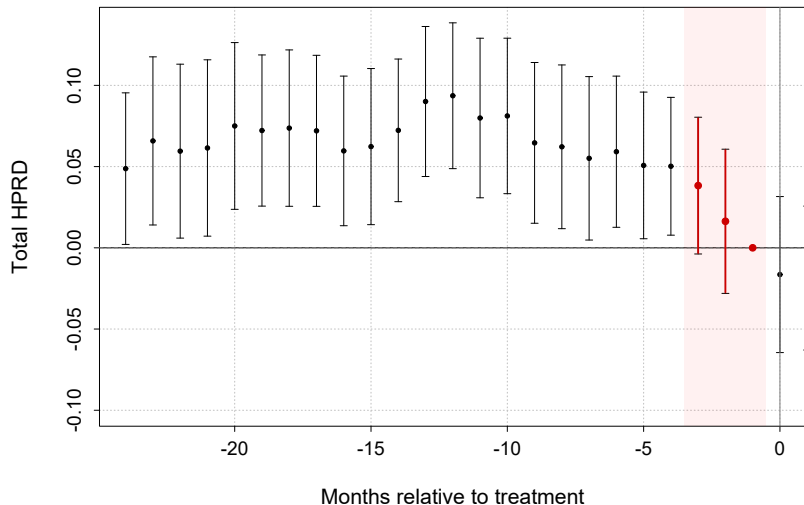
$$Y_{it} = \beta Post_{it} + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- ▶ $Post_{it} = 1$ in all months after a change
- ▶ β : **average post-change effect** on staffing Y_{it}
- ▶ Same fixed effects and controls as the event-study specification
- ▶ SEs clustered at the facility and month level

Identifying Assumptions

- **Parallel trends**
- **Staffing may change in anticipation of a formal ownership change**
 - ▶ A contract is signed, then there is a gap until the closing/transfer date.
 - ▶ During this gap, staffing may adjust
 - ▶ We want to observe the date the contract was signed.
- **Implication:** The months in this window may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period

Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \beta_\tau = 0 \quad \forall \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

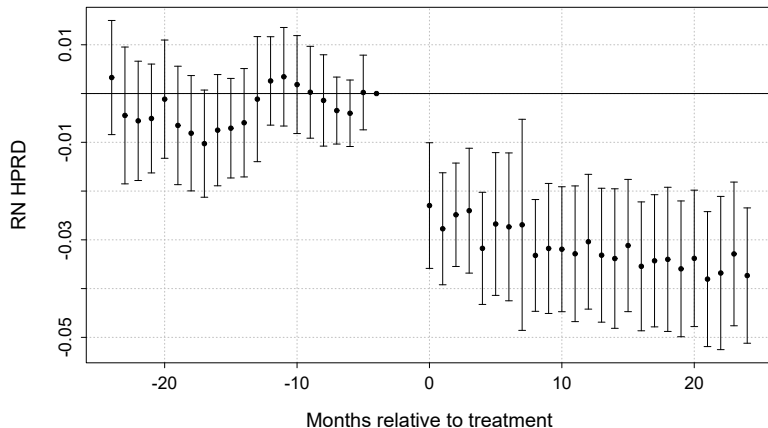
Table: Pre-trend Wald Test p-values (Total Staffing)

	Total (Levels)	Total (Logs)
2 Year Full Pre-Window	0.0006	0.0012
2 Year Window with Donut	0.2046	0.3268
1 Year Window with Donut	0.6387	0.6102

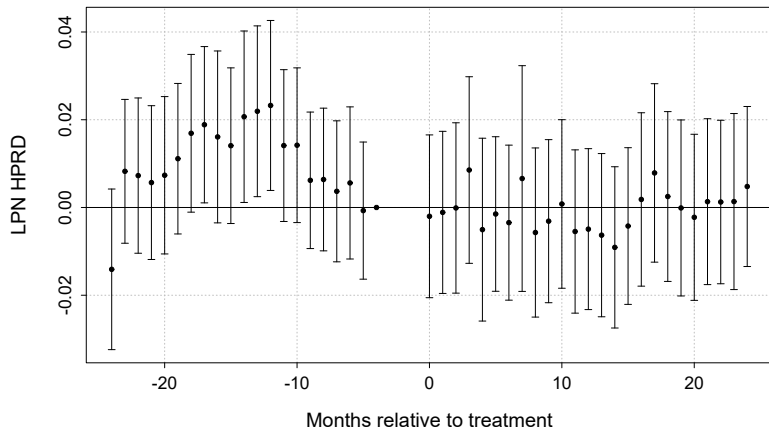
Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report p-value.

Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

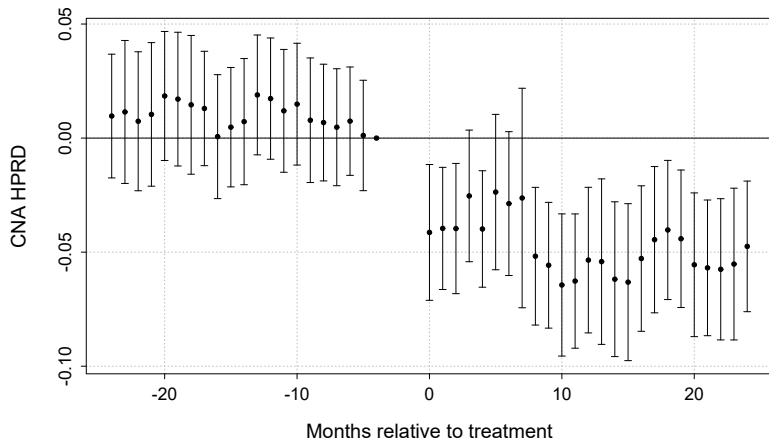
Event Study Results (RN HPRD)



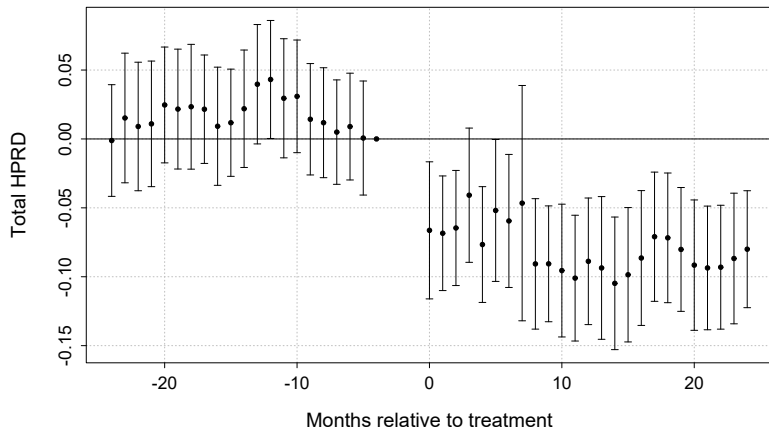
Event Study Results (LPN HPRD)



Event Study Results (CNA HPRD)



Event Study Results (Total Nursing HPRD)



TWFE DiD — Baseline (RN)

	RN	LPN	CNA	Total
HPRD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPRD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628,107$

TWFE DiD — Baseline (LPN)

	RN	LPN	CNA	Total
HPRD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPRD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628,107$

TWFE DiD — Baseline (CNA)

	RN	LPN	CNA	Total
HPRD	-0.032*** (0.005)	0.000 (0.006)	-0.054*** (0.009)	-0.086*** (0.013)
Log(HPRD)	-0.092*** (0.016)	0.003 (0.009)	-0.028*** (0.005)	-0.026*** (0.004)

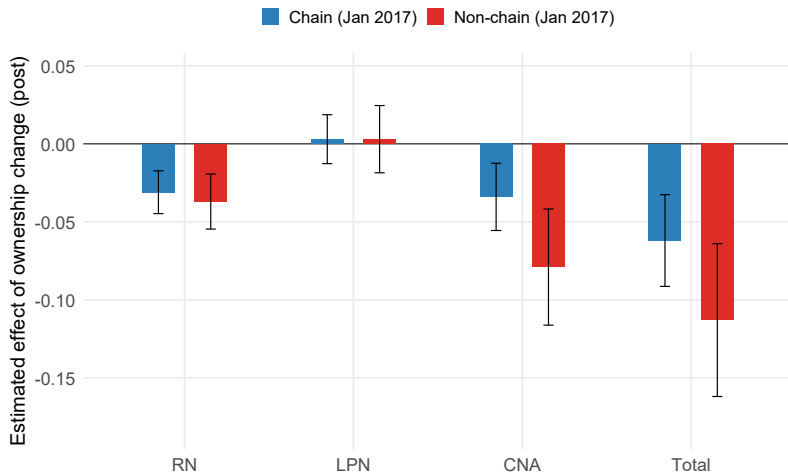
Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628,107$

TWFE DiD — Baseline (Total)

	RN	LPN	CNA	Total
HPRD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPRD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

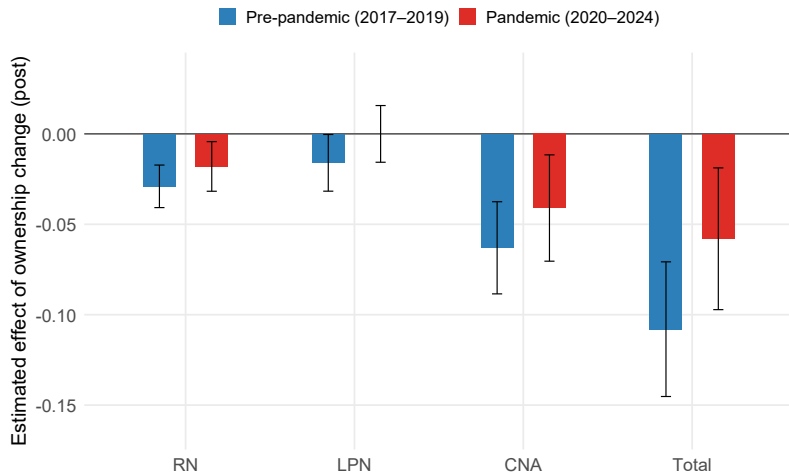
Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628,107$

Heterogeneity: Chain vs Non-Chain Facilities



Staffing declines are larger for non-chain facilities, nearly double for CNA and Total.

Heterogeneity: Pre vs Post-Pandemic



Large pre-pandemic declines, and post-pandemic effects are almost cut in half.

Robustness

- Stacked Difference-in-Differences
- Smaller and wider Donut Hole
- Gaps in the Data

Key Findings

- **Staffing falls after ownership change, but not uniformly.**

Key Findings

- **Staffing falls after ownership change, but not uniformly.**
 - ▶ Declines in **RN**, **CNA**, and **Total** HPRD

Key Findings

- **Staffing falls after ownership change, but not uniformly.**
 - ▶ Declines in **RN**, **CNA**, and **Total** HPRD
 - ▶ **No meaningful change** in **LPN** staffing

Key Findings

- **Staffing falls after ownership change, but not uniformly.**
 - ▶ Declines in **RN**, **CNA**, and **Total HPRD**
 - ▶ **No meaningful change** in **LPN** staffing
- **Magnitudes are relatively small.**

Key Findings

- **Staffing falls after ownership change, but not uniformly.**
 - ▶ Declines in **RN**, **CNA**, and **Total** HPRD
 - ▶ **No meaningful change** in **LPN** staffing
- **Magnitudes are relatively small.**
 - ▶ Effects are between 2 and 9%

Key Findings

- **Staffing falls after ownership change, but not uniformly.**
 - ▶ Declines in **RN**, **CNA**, and **Total HPRD**
 - ▶ **No meaningful change** in **LPN** staffing
- **Magnitudes are relatively small.**
 - ▶ Effects are between 2 and 9%
- **Effects are heterogeneous across settings.**

Key Findings

- **Staffing falls after ownership change, but not uniformly.**
 - ▶ Declines in **RN**, **CNA**, and **Total HPRD**
 - ▶ **No meaningful change** in **LPN** staffing
- **Magnitudes are relatively small.**
 - ▶ Effects are between 2 and 9%
- **Effects are heterogeneous across settings.**
 - ▶ Larger declines **pre-pandemic** vs. **pandemic period**

Key Findings

- **Staffing falls after ownership change, but not uniformly.**
 - ▶ Declines in **RN**, **CNA**, and **Total HPRD**
 - ▶ **No meaningful change** in **LPN** staffing
- **Magnitudes are relatively small.**
 - ▶ Effects are between 2 and 9%
- **Effects are heterogeneous across settings.**
 - ▶ Larger declines **pre-pandemic** vs. **pandemic period**
 - ▶ Larger declines for **non-chain** vs. **chain** facilities

Discussion

- **Key finding:** RN, CNA, and Total hours fall after ownership change; **LPN hours do not.**
- **Why might LPN hours not decline?**
 - ▶ Compliance
 - ▶ Cost
 - ▶ Substitution

Discussion

- **Question:** Are our findings economically **meaningful**?

Staff type	Sample Average	Effects
RN	25.6 min/day	−1.9 min/day
LPN	47.8 min/day	0.0 min/day
CNA	2 hr 6 min/day	−3.2 min/day
Total	3 hr 19 min/day	−5.2 min/day

- Increasing RN HPRD by 0.3 ($\approx$ 18 minutes) improves deficiencies 16% [Lin \(2014\)](#).
 - ▶ Our results would translate to a 1.6% increase in deficiencies.

Discussion

- **Heterogeneous effects**

- ▶ Staffing declines are **larger in non-chain facilities**
- ▶ Staffing declines are **smaller after the Covid-19 period**

Discussion

- **Heterogeneous effects**

- ▶ Staffing declines are **larger in non-chain facilities**
- ▶ Staffing declines are **smaller after the Covid-19 period**

- **Why are declines larger in non-chain facilities?**

- ▶ Small business
- ▶ Local

Discussion

- **Heterogeneous effects**

- ▶ Staffing declines are **larger in non-chain facilities**
- ▶ Staffing declines are **smaller after the Covid-19 period**

- **Why are declines larger in non-chain facilities?**

- ▶ Small business
- ▶ Local

- **Why are declines smaller after Covid?**

- ▶ Labor market shocks
- ▶ Higher oversight/compliance

Conclusion and future extensions

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN staffing remains stable**

Conclusion and future extensions

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN staffing remains stable**
- Declines are small

Conclusion and future extensions

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN staffing remains stable**
- Declines are small
- Effects larger for **non-chains** and smaller during **COVID**

Conclusion and future extensions

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN staffing remains stable**
- Declines are small
- Effects larger for **non-chains** and smaller during **COVID**
- Takeaway: ownership changes affect care via **internal production decisions**, even without changing market-structure

Conclusion and future extensions

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN staffing remains stable**
- Declines are small
- Effects larger for **non-chains** and smaller during **COVID**
- Takeaway: ownership changes affect care via **internal production decisions**, even without changing market-structure
- Future Research

Conclusion and future extensions

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN staffing remains stable**
- Declines are small
- Effects larger for **non-chains** and smaller during **COVID**
- Takeaway: ownership changes affect care via **internal production decisions**, even without changing market-structure
- Future Research
 - ▶ Linking our findings to quality metrics and resident outcomes

Conclusion and future extensions

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN staffing remains stable**
- Declines are small
- Effects larger for **non-chains** and smaller during **COVID**
- Takeaway: ownership changes affect care via **internal production decisions**, even without changing market-structure
- Future Research
 - ▶ Linking our findings to quality metrics and resident outcomes
 - ▶ Identifying mechanisms

Thank you

Questions?

References I

- Bowblis, J. R. and Brunt, C. S. (2025). Nursing home staffing expenditures and levels are impaired by high medicaid payer-mix. *Journal of the American Medical Directors Association*, 26(8):105723.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Gupta, A., Howell, S. T., Yannelis, C., and Gupta, A. (2021). Owner incentives and performance in healthcare: Private equity investment in nursing homes. Working Paper 28474, National Bureau of Economic Research.
- Harrington, C., Zimmerman, D., Karon, S. L., Robinson, J., and Beutel, P. (2000). Nursing home staffing and its relationship to deficiencies. *The Journals of Gerontology: Series B*, 55(5):S278–S287.
- Holmes, J. S. (1996). The effects of ownership and ownership change on nursing home industry costs. *Health Services Research*, 31(3):327–346.

References II

- Lin, H. (2014). Revisiting the relationship between nurse staffing and quality of care in nursing homes: An instrumental variables approach. *Journal of Health Economics*, 37:13–24.
- Park, J. and Werner, R. M. (2011). Changes in the relationship between nursing home financial performance and quality of care under public reporting. *Health Economics*, 20(7):783–801.